

Program Name: Firefly Motor

Motor Code: ML2A05A

Motor Model: ML2A

Serial No: 076

DATASHEET

PARAMETER	GOLDEN SAMPLE	DUT	DEV (%)
Model	ML2A05A-51V_P05	—	—
Nominal DC Voltage	51.00 V	51.96 V	+1.88%
Peak Power	6.58 kW	6.30 kW	-4.32%
Peak Torque	45.60 Nm	45.94 Nm	+0.75%
Peak Current AC	240.00 A	237.43 A	-1.07%
Motor Max Speed	7200 RPM	7200 RPM	+0.00%

PDI (ELECTRICAL)

TEST	ACCEPTANCE	DUT VALUE	STATUS
Hipot Average	< 3.50 mA	1.00	OK
IR Average	≥ 10.00 MΩ	13.33	OK
Resistance Average	≈ 15.30 mΩ	15.00	OK

BACK-EMF ANALYSIS

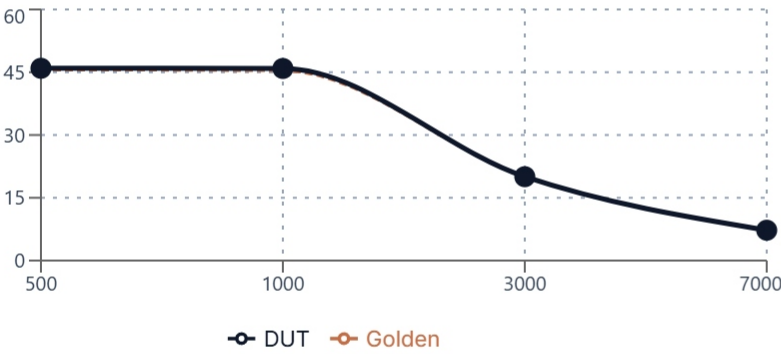
PARAMETER	GOLDEN (V)	DUT (V)	DEV (%)
Avg BEMF @ 3000 RPM	28.50	28.15	-1.23%

POSITION SENSOR PARAMETERS

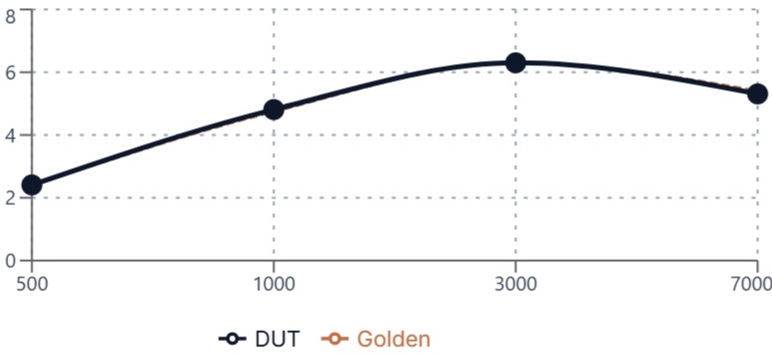
PARAMETER	GOLDEN SAMPLE	DUT	STATUS
Sine Min	1150	1150	OK
Sine Max	2900	2900	OK
Cos Min	1150	1150	OK
Cos Max	2900	2900	OK
Sine P-P	1750	1750	OK
Cos P-P	1750	1750	OK
Offset	766	766	OK

PERFORMANCE CURVES — GOLDEN SAMPLE VS DUT

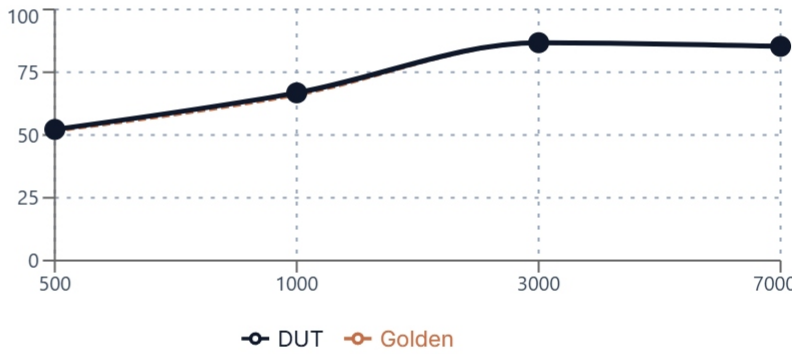
TORQUE vs SPEED (Nm)



POWER vs SPEED (kW)



GROUP EFF vs SPEED (%)



06 // TORQUE VS SPEED

SPEED (RPM)	GOLDEN (NM)	DUT (NM)	DEV (%)
500	45.57	45.94	+0.81%
1000	45.37	45.90	+1.17%
3000	20.06	20.04	-0.10%
7000	7.37	7.25	-1.63%

07 // POWER VS SPEED

SPEED (RPM)	GOLDEN (KW)	DUT (KW)	DEV (%)
500	2.39	2.41	+0.64%
1000	4.75	4.81	+1.19%
3000	6.30	6.30	-0.07%
7000	5.40	5.31	-1.58%

08 // EFFICIENCY VS SPEED

SPEED (RPM)	GOLDEN (%)	DUT (%)	DEV (%)
500	51.50	52.22	+1.40%
1000	65.87	66.79	+1.40%
3000	86.51	86.76	+0.29%
7000	85.67	85.31	-0.42%

END OF LINE RESULT

PASSED

Tested By:
GGS
2026-06-10

Checked By:
HRN
2026-06-10

Approved by:
IMR
2026-06-10